REFLOWECONOMY · REGULATORY AND STANDARDS REGISTER

ReflowEconomy regulatory and standards register

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First register (kit 1.8 pipeline) for the reference micro-factory, with the first partner region (Maharashtra, India) and the international standards the design uses

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Which standards and regulations apply to a small plastics micro-factory, where, and what is needed. The first partner region is Maharashtra, India (first candidate partner SWaCH, Pune; nothing agreed), so state and national rules are listed for it; other regions are to assess when a partner is found. Not legal advice; anything marked "to assess" needs a qualified reviewer before any site operates or any claim is made. Method: STANDARDS section 23. Every source was checked on 2026-10-25.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Plastic Waste Management Rules, 2016, as amended (registration of plastic waste processors)	India; administered in Maharashtra by MPCB	Processors register on the CPCB portal (eprplastic.cpcb.gov.in); fresh registration valid one year, renewal three years; application fee by capacity, INR 5,000 under 200 t a year; physical audit by the board (MPCB service page)	Applies: the reference site processes about 25 t a year (PR-020)	Site operator	Applies; to do at P5 with a real site
G2	Air (Prevention and Control of Pollution) Act, 1981 and Water (Prevention and Control of Pollution) Act, 1974: consents; Hazardous and Other Wastes Rules, 2016: authorisation	India	MPCB requires the Air and Water Act consents and the hazardous waste authorisation as part of processor registration (same page)	Applies: melt fumes discharged through a stack; wash water and sludge; the industry category and consent route for a 25 t a year site are to assess	Site operator with a consultant	To assess
G3	IS 14534:1998, *Guidelines for recycling of plastics*	India (BIS)	Selection, segregation and processing; wastewater treated before discharge (7.3.1); no burning of plastic waste (8.4); indoor air quality and exhausts (8.2); fire protection (8.3); no recycled material in food, pharmaceutical or drinking-water contact (8.5) (BIS text)	Applies as guidance; the design meets 7.3.1, 8.2, 8.3 and 8.4 by design; 8.5 goes into the passport and product rules	Design lead	Met by design (8.5 to add to the playbook)
G4	Solid Waste Management Rules, 2016, Rule 15	India	Local bodies recognise organisations of waste pickers and integrate them into collection	Applies to the partner model, not the hardware	Partner and municipality	Applies
G5	Occupational Safety, Health and Working Conditions Code, 2020 (in force 21 November 2025; replaces the Factories Act, 1948)	India	A factory is a workplace of 20 or more workers with power (factoHR summary); under the Factories Act the threshold was 10 (Factories Act, s. 2(m))	A four-worker site is below the factory threshold; which of the code's general duties still apply, and the state rules, are to assess	Site operator	To assess

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G6	MSEDCL Conditions of Supply, clause 6.4.1	Maharashtra	Single-phase supply up to 40 A; above that, three-phase LT up to 80 kW (MSEDCL)	Applies: the morning loads exceed 40 A on paper (RFE-DEC-001 D1)	Site operator and electrician	Applies; decision with Amish
G7	E-Waste (Management) Rules, 2022 and Battery Waste Management Rules, 2022	India	Boards and cells go only to registered recyclers or refurbishers	Applies to the export cage and the cell container; receiver types to name per stream (PR-004)	Site operator	Applies; per-stream rules to write
G8	ISO 13857:2019, safety distances to prevent hazard zones being reached	International	Table 4: reaching through slots of 40 to 120 mm, 850 mm; over 120 mm, the reaching-over distances (Troax; Axelent)	Applies to the shredder feed slot (460 x 120 mm, 1,060 mm path)	Design lead	Met by design (RFE-CAL-001 section 7; RFE-DEC-001 D3)
G8a	EN 12012 series, safety requirements for size reduction machines for plastics and rubber (type C)	European; used internationally	Machine-specific requirements that take precedence over the generic ISO 13857 distances	Applies to the shredder; which part applies and its feed-opening rules are to assess (text not checked)	Design lead and safety professional	To assess
G9	ISO 14119 (interlocking devices) and ISO 13849-1 (safety-related parts of control systems)	International	Interlock selection and the performance level of the feed flap, door and heater interlocks	Applies to the shredder interlocks and the heater interlock	Electrician and safety professional	To assess (performance level not yet set)
G10	IEC 60204-1, electrical equipment of machines; the local wiring code	International; local	Emergency stop, isolation, protective bonding of the machines and the board	Applies to the board and the machines	Licensed electrician	Certification needed later (installation certificate, build plan stop S2)
G11	IEC 60034-30-1, efficiency classes of motors	International	Minimum motor efficiencies used in the energy budget (ABB note)	Applies to the motors bought (IE3 assumed)	Design lead	Met by design (on paper)
G12	Occupational exposure limits (local; the playbook uses	Local; United States as a floor	Air at the workstations	Applies; can only be shown by air	Safety professional	To assess at TRL 4 (on hold)

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	the US OSHA PELs as a floor: formaldehyde 0.75 ppm over 8 h, acrolein 0.1 ppm)			monitoring (PR-011)		
G13	Noise at work (local limit; the playbook uses 85 dB(A) over 8 h, as in EU Directive 2003/10/EC)	Local	Operator exposure and hearing protection zones	Applies (PR-012)	Safety professional	To assess (measurement at TRL 4)
G14	National building and fire code (in India, the National Building Code 2016, Part 4)	Local	Exits, escape widths, headroom, extinguishers, storage of combustible stock	Applies to the layout; the safety professional confirms escape widths and the 2.02 m headroom (stop S7)	Safety professional	To assess
G15	Container modification: structural design	Local	Removing the adjoining side walls of two containers	Applies to the container option only; the rented shed (preferred) needs none	Structural engineer	Certification needed later
G16	ISO 14021, self-declared environmental claims (recycled content)	International	Recycled content statements in passport schema v0.2	Applies to every passport	Design lead	Met by design (schema v0.2)
G17	Basel Convention plastic waste amendments (in force 1 January 2021)	International	Prior informed consent for most mixed or contaminated plastic waste shipments	Applies to export lots that cross borders; most export streams here are metals and boards to national refiners	Site operator	To assess per stream
G18	EU Regulations 2024/1781 (ESPR, digital product passport)	European Union	Product and battery passports	Direction of travel for the passport only; no compliance is claimed	Design lead	Does not apply (reference only)

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	and 2023/1542 (batteries)					

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

All checked on 2026-10-25: MPCB, *Plastic processor registration* service page; BIS, IS 14534:1998; factoHR, *OSH Code 2020* summary; Factories Act, 1948 (IRIMEE copy); MSEDCL, *Conditions of Supply*, 2010; Troax, *Safety guide*; Axelent, *Standards*; ABB, *Technical note IEC 60034-30-1*, 2018. The OSHA limits, ISO 14021, Basel Convention and EU regulations are cited with links in RFE-PRB-001 and RFE-CAL-001.